Scalable Fraud Risk Triage & Analytics System

Detecting Fraud Patterns, Prioritizing High-Risk Transactions and Generating AI-Driven Risk Intelligence.

SOURCE: Synthetic Financial Transaction Datasets (Fraud Simulation Dataset)

View Fraud Risk Overview

11 Mar 2026, 04:22 AM

Prepared by **Navina M** | Fraud Risk & Trust Analytics

Fraud Risk Overview

Fraud Monitoring & Risk Metrics – 2024

Country: All
Month: All
Device: All
Category: All

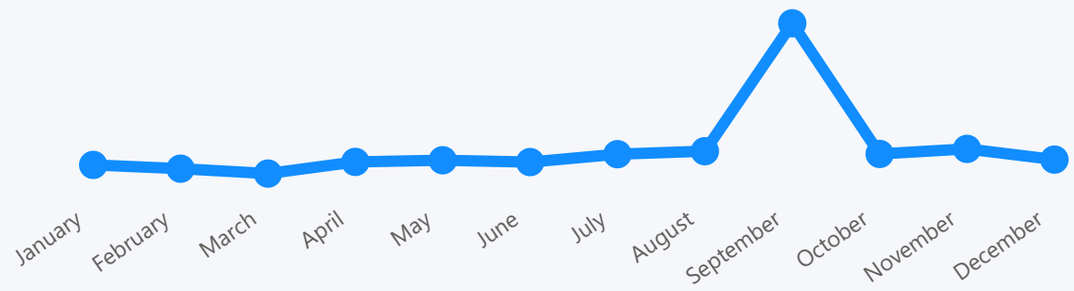
Total Transactions
800K

Fraud Transactions
24.1K

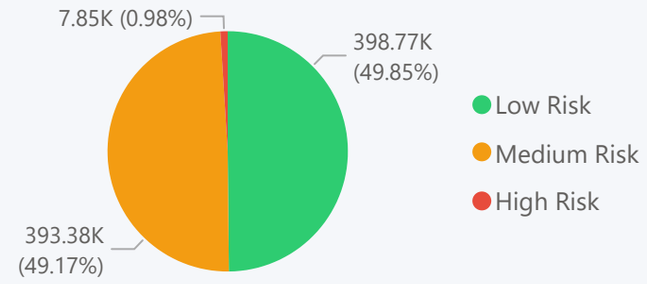
High Risk Transactions
7.8K

Fraud Rate
3.01%

Monthly Fraud Rate Trend

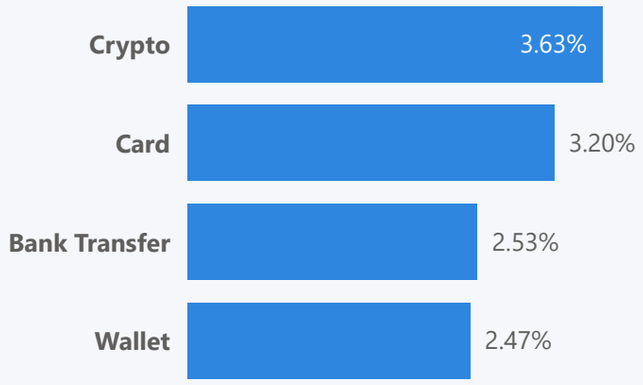


Risk Score Distribution



Fraud Transactions by Country

Fraud Rate by Payment Method



Fraud Investigation & Risk Analysis

High-Risk Transaction Monitoring and Fraud Pattern Analysis

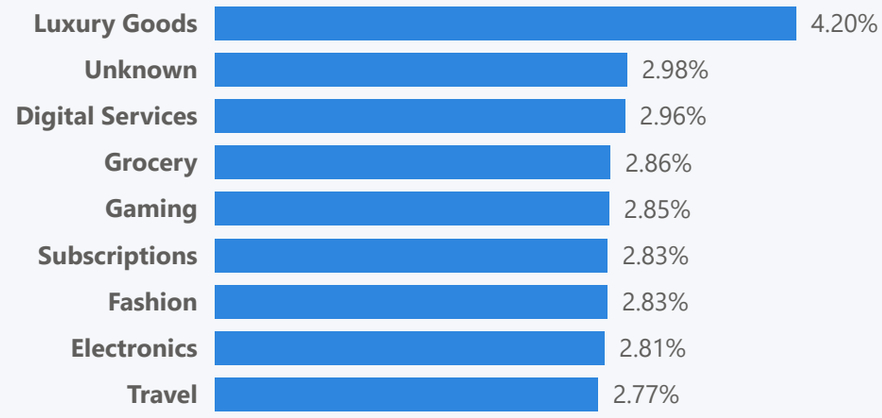
Country: All | Month: All | Device: All | Payment Method: All

🚨 CRITICAL FRAUD ALERT DETECTED
14 High Risk Transactions Require Investigation

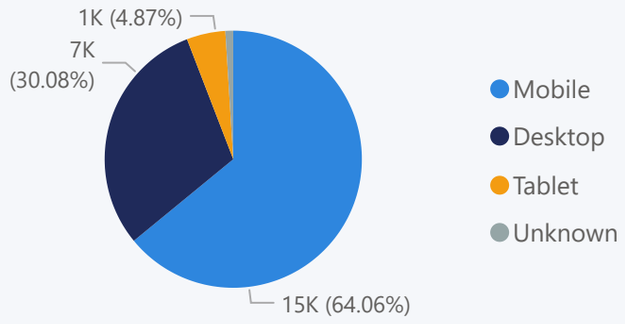
High Risk Transaction Investigation Queue

Transaction ID	User ID	Amount	Payment Method	Country	Risk score	Fraud Attack Pattern
674800	20434	702.38 INR	Crypto	Vietnam	100	Crypto Payment Fraud
357801	33615	509.52 INR	Crypto	Vietnam	90	Crypto Payment Fraud
350418	38165	525.32 GBP	Crypto	Nigeria	90	Crypto Payment Fraud
715547	13478	526.72 USD	Crypto	Nigeria	90	Crypto Payment Fraud
188192	3547	537.56 USD	Crypto	Indonesia	90	Crypto Payment Fraud

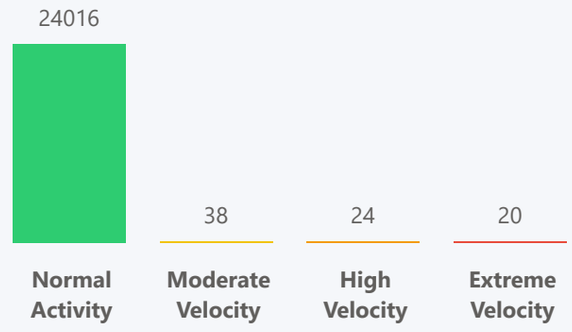
Fraud Risk by Merchant Category



Fraud Distribution by Device

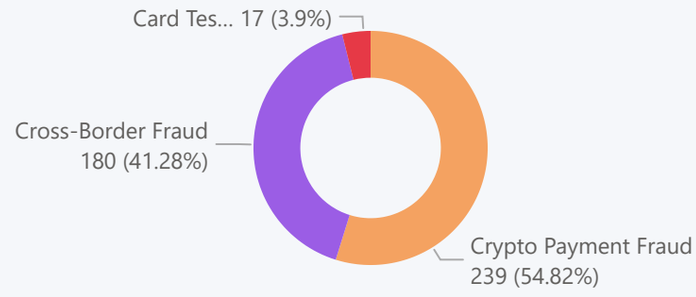


Transaction Velocity Risk Levels



Detected Fraud Attack Patterns

(High-Risk Activity)



AI Fraud Intelligence

AI-Powered Fraud Pattern Analysis & Risk Insights

Total Transactions	High Risk Transactions	Fraud Exposure
800K	7.8K	4.58M

🔎 Fraud Intelligence Summary

KEY FRAUD INSIGHTS

- Overall fraud rate stands at **3.01%** across **800K transactions**, indicating moderate but persistent fraud activity.
- **Crypto transactions** present the highest fraud risk among payment methods.
- Fraud activity shows a noticeable **spike in September**, suggesting possible coordinated attack periods.

RISK DRIVERS

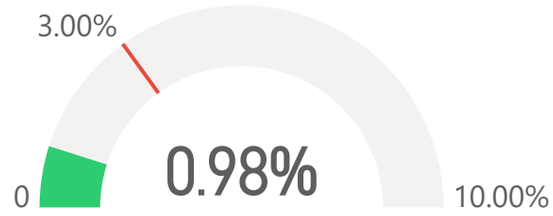
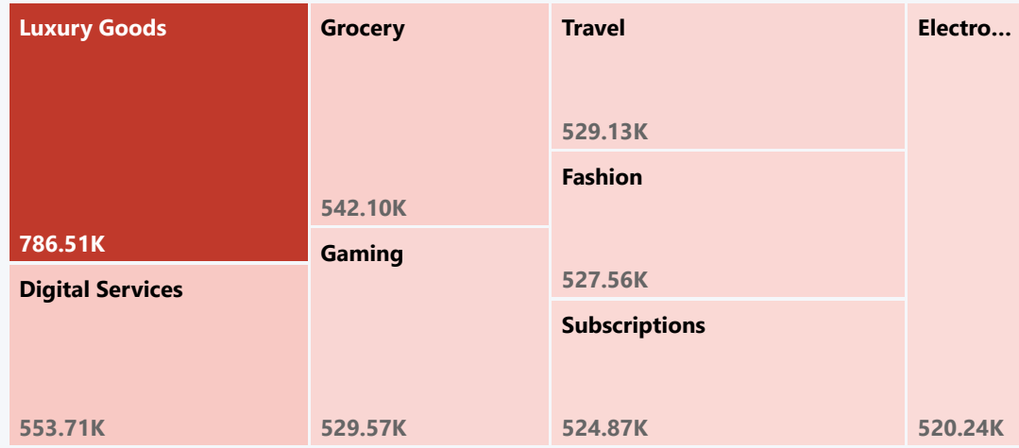
- **Luxury Goods merchants** demonstrate the highest fraud vulnerability.
- **Mobile devices** account for the majority of suspicious transactions.
- Fraud patterns are primarily driven by **Crypto Payment Fraud** and **Cross-Border Fraud**.

RECOMMENDATIONS

- Introduce enhanced authentication and monitoring for **crypto transactions**.
- Implement stricter controls for **high-value luxury goods purchases**.
- Investigate the **September fraud spike** to refine detection models and velocity rules.

High-Risk Transaction Concentration

Operational Risk Threshold: 3%



Fraud Exposure by Merchant Cateogry